

Software Quality Assurance Principles

This is the first page of Sample Document 2. It is used as the secondary upload file in Merge PDF tests that require two or more documents, and as the second file in multi-file upload boundary tests.

Quality assurance in modern web applications requires a multi-layered approach: unit tests validate individual functions, integration tests verify component interactions, and end-to-end tests confirm complete user journeys from browser to backend and back.

The iLovePDF tool suite provides twelve distinct PDF manipulation features. Each feature follows the same user journey pattern: navigate → upload → configure → process → download. This consistent pattern makes the Page Object Model particularly effective here.

Data Table

Principle	Description	Priority	Coverage
Correctness	Tool produces valid output	Critical	E2E
Usability	UI elements accessible	High	Regression
Reliability	Consistent results	High	Smoke
Performance	Reasonable process time	Medium	E2E

Selenium WebDriver Best Practices

Effective Selenium automation relies on robust element location strategies. This framework prioritises ID and CSS selectors over XPath where possible, falling back to attribute-based XPath (e.g., href-contains) only when stable IDs are unavailable.

Explicit waits via WaitUtil replace implicit waits for dynamic elements. The WaitStrategy enum (VISIBLE, CLICKABLE, PRESENCE, INVISIBLE) documents intent clearly and prevents over-waiting on static elements.

File upload is handled through direct sendKeys() to the hidden file input rather than simulating native OS dialogs, making tests headless-compatible and cross-platform without additional tooling.

Data Table

Strategy	Locator Type	Use Case	Stability
Primary	By.id	Form inputs, buttons	High
Secondary	By.cssSelector	Style-based	High
Fallback	By.xpath (href)	Navigation tiles	Medium
Avoid	By.xpath (text)	Localisation risk	Low

Defect Lifecycle & Reporting

When a test fails, the TestListener captures a timestamped screenshot and attaches it to the ExtentReports entry. The failure message, stack trace, and TC ID are all recorded, enabling rapid root-cause analysis.

Defects identified by automation follow the standard lifecycle: New → Assigned → In Progress → Fixed → Verified → Closed. The TC ID in each assertion message provides a direct link between the automated failure and the corresponding manual test case in the test register.

The RetryAnalyzer retries flaky tests up to two times before marking them as failed. Tests that consistently fail after retries are promoted to high-priority defects and assigned immediately.

Data Table

Status	Owner	Action	SLA
New	Automation	Log & screenshot	Immediate
Assigned	Dev Lead	Triage	4 hours
In Progress	Developer	Fix	2 days
Verified	QA	Re-run test	1 day